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Stat 50

HW 13#

(1)

(a) $P(Z > -0.85) = 1 - P(Z \leq -0.85)$

$= 1 - 0.1977 = 0.8023$

(b) $P(0.40 \leq Z \leq 1.30) = P(Z \leq 1.30) - P(Z \leq 0.40)$

$= 0.9032 - 0.6554$

$= 0.2478$

(c) $P(-0.30 \leq Z \leq 0.90) = P(Z \leq 0.90) - P(Z \leq -0.80)$

$= 0.8159 - 0.3821$

$= 0.4338$

(d) $1 - P(-1.50 \leq Z \leq -0.45) = 1 - P(Z \leq -0.45) + P(Z \leq -1.5)$

$= 1 - 0.3264 + 0.0668 = 0.7404$

(3) $Z \sim N(0,1)$

(a) $P(Z \geq C) = 0.1587$

$0.1587 = P(Z \leq -1.0) \Rightarrow P(Z \geq 1.0)$

$\therefore$ $C = 1.00$

(b) $P(C \leq Z \leq 0) = 0.4772$

$P(Z \leq 0.00) - P(Z \leq C) = 0.4772$

$\Rightarrow P(Z \leq C) = P(Z \leq 0.00) - 0.4772$

(c)

$\Rightarrow P(Z \leq C) = 0.5000 - 0.4772$

$P(Z \leq C) = 0.0228$

$\boxed{C = -2.00}$

(e) $P(-C \leq Z \leq C) = 0.8664$

$1 - 0.8664 = 0.1336$

$0.1336/2 = 0.0668$

then $\boxed{C = 1.50}$

(d) $P[0 \leq Z \leq C) = 0.2967$

$\Rightarrow P(Z \leq C) - P(Z \leq 0) = 0.2967$

$\Rightarrow P(Z \leq C) = 0.2967 + 0.5000$

$\Rightarrow P(Z \leq C) = 0.7967$

$\Rightarrow \boxed{C = 0.83}$

(e) $P[|Z| \geq C) = 0.1470$

$\Rightarrow P[|Z| \leq C) = 1 - 0.1470 = 0.853$

$\boxed{C = 1.45}$

(5)

$\mu = 25.1$

$\sigma = 0.08$

(a) $x \sim N(25.1, (0.08)^2)$

$P(x \leq 25.0) = P\left(\frac{x - 25.1}{\sqrt{(0.08)^2}} < \frac{25.0 - 25.1}{0.08}\right)$

$\Phi(Z < -1.25) = 0.1056$

(b)

$P(x \leq c) = 0.1000 \Rightarrow c = -1.28$

$\Rightarrow -1.28 = \frac{x - 25.1}{0.08} \Rightarrow x = (-1.28)(0.08) + 25.1$

$x = 24.997 \text{ mm}$

(c)

$\frac{x - 25.1}{0.08} = c$

$\Rightarrow \frac{25.2 - 25.1}{0.08} = c$

$\Rightarrow 1.25 = c \qquad P(Z \leq 1.25) = 0.8944$

then, 89th percentile

(d) $P(25.0 < X < 25.3)$

$= P\left(\frac{25.0 - 25.1}{0.08} < \frac{X - 25.1}{0.08} < \frac{25.3 - 25.1}{0.08}\right)$

$= P(-1.25 < Z < 2.5) = P(Z \leq 2.5) - P(Z \leq -1.25)$

$= 0.9938 - 0.1056 = 0.8882$

(6)

$\mu = 822 \ \mu m$

$\sigma = 29 \ \mu m$

(a) $P(X \leq c) = .1000$

$c = -1.28$

$P\left(\frac{X - 822}{29} \leq \frac{-1.28 - 822}{29}\right)$

$\Rightarrow -1.28 = \frac{x - 822}{29}$

$\Rightarrow \boxed{x = 784.88}$

(5) $780 \ \mu m$

$P(x \leq 780) = P\left(\frac{x - 822}{29} \leq \frac{780 - 822}{29}\right)$

$P(Z \leq -1.45) = 0.0735$

$\therefore 7^{th}$ percentile

(5)

(c)

$$P(800 \le x \le 830) = P\left(\frac{800-822}{27} \le \frac{x-822}{27} \le \frac{830-822}{27}\right)$$

$$= P(-0.76 \le Z \le 0.28)$$

$$= P(Z \le 0.28) - P(Z \le -0.76)$$

$$= 0.6108 - 0.2236$$

$$= 0.3867$$

(9)

$$\mu = 1400 \qquad , \sigma = 200$$

(a) $P(x > 1800) = 1 - P(x \le 1800)$

$$= 1 - P\left(\frac{x-1400}{200} \le \frac{1800-1400}{200}\right)$$

$$= 1 - P(Z \le 2.00)$$

$$= 0.0228$$

(b) $P(x \le c) = .1000 \Rightarrow c = -1.28$

$$\Rightarrow -1.28 = \frac{x - 1400}{200} \Rightarrow x = (-1.28(200)) + 1400$$

$$x = 1144$$

(c)

(b)

(c)

$$\frac{x - 1400}{200} = \frac{1645 - 1400}{200}$$

$$P(x \leq 1.225) = 0.8888$$

88th percentile

(d) $P(1350 \leq x \leq 1550)$

$$= P\left(\frac{1350 - 1400}{200} \leq \frac{x - 1400}{200} \leq \frac{1550 - 1400}{200}\right)$$

$$= P(-0.25 \leq z \leq 0.75)$$

$$= P(z \leq 0.75) - P(z \leq -0.25)$$

$$= 0.7734 - 0.4013$$

$$= 0.3721$$

(e)

(10) $\mu = 65.0$, $\sigma = 100$

No, the maximum score is 800 which is $1.5(100) + 650$ and there are no score after that. There are no possibility that score of these freshmen are normally distributed.

(13)

(a) $\mu_x = 15$ cm , $\sigma_x = 0.025$ cm

$\mu_y = 14.88$ cm , $\sigma_y = 0.015$ cm

$C = \frac{1}{2}(X - Y) = \frac{1}{2}X - \frac{1}{2}Y$

$\mu_C = \left(\frac{1}{2}\right)\mu_X - \left(\frac{1}{2}\right)\mu_Y = \left(\frac{1}{2}\right)(15) + \left(-\frac{1}{2}\right)(14.88)$

$\mu_C = 0.06$

(b) $\sigma_C = \sqrt{\left(\frac{1}{2}\right)^2 (0.025)^2 + \left(\frac{1}{2}\right)^2 (0.015)^2}$

$\sigma_C = 0.01458$ cm

(c) $P(X \leq 0.05) = P\left(\frac{X - 0.06}{0.01458} < \frac{0.05 - 0.06}{0.01458}\right)$

$P(Z < -0.69) = 0.2451$

(d) $P(x \leq c) = 0.2500$

$c = -0.67$

$\frac{x - 0.06}{0.01458} = -0.67$

$x = 0.0502$

(e) $P(0.05 < x < 0.09)$

$= P\left(\frac{0.05 - 0.06}{0.01458} < x < \frac{0.09 - 0.06}{0.01456}\right)$

$= P(-0.69 < z < 2.06)$

$= P(z < 2.06) - P(z < -0.69)$

$= 0.7552$

(f)

(9)

(15)

$\mu = 12.05$, $\sigma = 0.03$

(a) $P(x < 12) = P\left(Z \leq \frac{-0.05}{0.03}\right)$

$P(Z \leq -1.67) = 0.0475$

(b) $P(x \leq c) = .99$

$.99 = \frac{x - 12.05}{0.03}$

$x = (.99)(0.03) + 12.05$

$\boxed{x = 12.07}$

(c)

$\mu = 12.05$

$\boxed{Z = -2.23}$

$\Rightarrow \sigma = \frac{x - \mu}{Z} = \frac{12 - 12.05}{-2.33}$

$\boxed{\sigma = 0.02150}$

(23)

(a) $R_1 \sim N(100, 5^2)$

$R_2 \sim N(120, 10^2)$

(10)

$P(R_2 > R_1) = P(R_2 - R_1 > 0)$

$R_2 - R_1 \sim N(120 - 100, 10^2 + 5^2)$

$R_2 - R_1 \sim N(20, 125)$

$P\left(\frac{R_2 - R_1 - 20}{\sqrt{125}} > \frac{-20}{\sqrt{125}}\right) = P(Z > -1.788)$

$= 1 - P(Z \leq -1.79)$

$= 1 - 0.0367 = 0.9633$

(b) $P(R_2 > R_1 + 30) = P(R_2 - R_1 > 30)$

$R_2 - R_1 \sim N(20, 125)$

$P\left(\frac{} > \frac{30 - 20}{\sqrt{125}}\right) = P(Z > 0.89)$

$= 1 - P(Z \leq 0.89) = 1 - 0.8133$

$= 0.1867$

(c) $P(R_1 + R_2 < 200) \Rightarrow R_1 + R_2 \sim N(220, 125)$

$P\left(Z < \frac{200 - 220}{\sqrt{125}}\right) = P(Z < -1.79)$

$= 0.0367$

25 $\varepsilon \sim (0, 0.25^2)$

(a) $P(X > 0.5) = P\left(Z > \frac{0.5 - 0}{0.25^2}\right)$

$= P(Z > 1) = 1 - P(Z \leq 1)$

$= 0.1587$

(b) $P(Z > z) = 0.01 \Rightarrow z = 2.33$